DEPARTMENT OF
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GEETHANJALI COLLEGE OF ENGINEERING AND TECHNOLOGY

Digital Communication Lab project

Topic: Digital Lock System

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ABSTRACT

The Digital Lock System is a security-based project developed to provide controlled access to restricted areas using a password entered through a keypad. The system operates on the principle of digital communication between input and output components, where all signals are transmitted and processed in digital form. The main controller of the system is an Arduino Nano, which receives input signals from the keypad and processes them to verify the correctness of the entered password.

When a valid password is entered, the Arduino sends a digital PWM signal to a servo motor, which rotates to simulate the unlocking of a door. However, if the entered password is incorrect, the system counts the number of wrong attempts. After three consecutive wrong passwords, the Arduino activates a buzzer through a digital HIGH signal and disables further input for 30 seconds, thereby preventing unauthorized access. This time delay acts as an additional layer of security to discourage repeated trial attempts.

This project highlights the importance of digital communication techniques in security systems, where components exchange information using binary signals (0s and 1s). The Digital Lock System provides a simple, cost-effective, and reliable solution for access control. It can be used in homes, offices, lockers, and labs to improve safety without the need for mechanical keys. Through this project, students gain practical knowledge about the use of microcontrollers and digital signal control in real-time applications.

CHAPTER 1
INTRODUCTION

In today's modern world, safety and security have become one of the most important requirements in both residential and industrial areas. Traditional lock and key mechanisms are slowly being replaced by electronic security systems that provide better protection and are more user-friendly. One such modern approach is the Digital Lock System, which uses electronic components to allow access only to authorized users through password verification.

The Digital Lock System using Arduino is a password-based security project designed to prevent unauthorized access. It uses a keypad to enter a password, a servo motor to represent the locking mechanism, and a buzzer to provide security alerts. The Arduino Nano acts as the central controller, processing the user input and performing the required actions based on the password entered.

When the user enters the correct password, the system unlocks the door by rotating the servo motor. If an incorrect password is entered, the system notifies the user through a buzzer. To increase the level of security, the system is programmed such that after three consecutive wrong password attempts, it automatically locks further access for 30 seconds and activates a buzzer alarm.

This project mainly demonstrates the concept of digital communication, as the information between the keypad, Arduino, buzzer, and servo motor is exchanged using digital signals. The system operates completely on binary logic — where each keypress, control signal, and output state is represented by either a HIGH (1) or LOW (0) signal.

The Digital Lock System is a practical example of how digital electronics and microcontroller-based systems can be combined to create an efficient and reliable security application. It is a simple, low-cost, and effective design that can be implemented in homes, offices, and other secure locations.

CHAPTER 2
COMPONENTS REQUIRED

The Digital Lock System using Arduino is built using simple electronic components that work together to form a secure password-based access control system. The components required for this project are listed below.

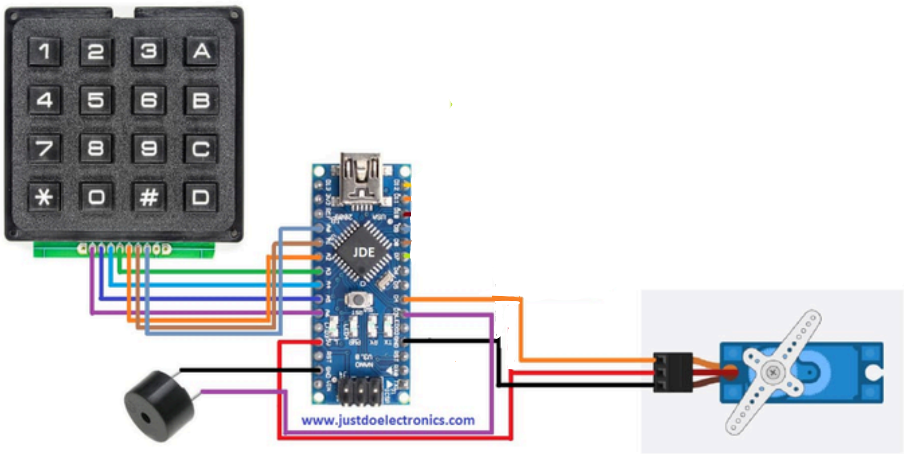
S.No	Component Name	Quantity	Description / Function
1	Arduino Nano	1	Acts as the main controller. It
2	4×3 Keypad	1	Used to enter the password. Each
3	Servo Motor (SG90)	1	Represents the door lock
4	Buzzer	1	Gives sound alert when a wrong
5	Breadboard	1	Used to make temporary
6	Jumper Wires (Male-to-Male)	As required	Used for making connections between Arduino, keypad, buzzer, and servo motor.
7	USB Cable / Power Source	1	Supplies power to the Arduino Nano

- ll input and output devices are connected to digital pins of the Arduino Nano.
- The servo motor requires a PWM digital pin to control its rotation.
- The buzzer works with digital HIGH/LOW signals to produce sound.
- The keypad works on a matrix digital input scanning principle.

CHAPTER 3: Circuit Diagram

The Digital Lock System circuit is designed using an Arduino Nano, a 4×4 keypad, a servo motor, and a buzzer. Each component is connected to the Arduino through digital pins to enable proper communication and control.

- **Circuit Description**



- The 4×4 Keypad is connected to Arduino's digital pins D2–D9. It is used for entering the password.
- The Servo Motor is connected to digital pin D10, which controls the lock movement (open or close).
- The Buzzer is connected to digital pin D11, which activates when the password is entered incorrectly three times.
- The Arduino is powered through USB or a 9V battery.
- All components share a common ground (GND) connection with the Arduino.

🔹 Pin Connections Table

Component	Arduino Pin
Keypad Rows (R1–R4)	D2–D5
Keypad Columns (C1–C4)	D6–D9
Servo Motor	D10
Buzzer	D11
5V Power	5V
GND	GND

🔹 Circuit Explanation

When the user enters a password on the keypad, the Arduino reads each key press. If the entered password matches the stored password, the servo motor rotates to unlock the door.

If the password is wrong three times in a row, the buzzer turns ON for alert, and the system locks input for 30 seconds before allowing further attempts.

CHAPTER 5
HARDWARE AND SOFTWARE IMPLEMENTATION

The Digital Lock System is implemented using both hardware and software components that work together to provide secure password-based access control.

- **Hardware Implementation**

The hardware part includes the Arduino Nano, keypad, servo motor, and buzzer.
Each component is connected using digital pins to send and receive signals between the Arduino and external devices.

Component	Arduino Pin
Keypad Rows (R1–R4)	D2–D5
Keypad Columns (C1–	D6–D9
Servo Motor	D10
Buzzer	D11
Power Supply	5V
Ground	GND

Hardware Working

The keypad is used to enter the password.
The Arduino checks if the entered password matches the stored one.
When the password is correct, the servo motor rotates to simulate unlocking.
If the password is wrong three times, the buzzer turns ON and the system locks for 30 seconds before allowing new attempts.

🔹 Software Implementation (Arduino Code)

The software is written and uploaded to the Arduino Nano using the Arduino IDE. It handles the logic for password checking, controlling the servo motor, and buzzer operation.

```
#include <Keypad.h>
#include <Servo.h>

// --- Servo setup ---
Servo lockServo;
int servoPin = 9;

// --- Buzzer setup ---
int buzzer = 10;

// --- Password setup ---
String password = "1121";  // Set your password here
String input = "";
int wrongAttempts = 0;

// --- Keypad setup ---
const byte ROWS = 4;
const byte COLS = 3;
char keys[ROWS][COLS] = {
  {'1','2','3'},
  {'4','5','6'},
  {'7','8','9'},
  {'*','0','#'}
};

byte rowPins[ROWS] = {2, 3, 4, 5}; // connect to R1, R2, R3, R4
byte colPins[COLS] = {6, 7, 8};    // connect to C1, C2, C3

Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS, COLS);

void setup() {
  Serial.begin(9600);
  lockServo.attach(servoPin);
  pinMode(buzzer, OUTPUT);
  lockServo.write(0);  // Locked position
  Serial.println("Enter Password:");
}

void loop() {
  char key = keypad.getKey();

  if (key) {
    if (key == '#') {
      checkPassword();
      input = ""; // clear after check
    }
    else if (key == '*') {

      input = ""; // clear input manually
      Serial.println("Cleared input");
    }
    else {
      input += key;
      Serial.print("*"); // hide actual password
    }
  }
}
```

```
// --- Check entered password ---
void checkPassword() {
  Serial.println();
  if (input == password) {
    Serial.println("Access Granted ✅");
    unlockDoor();
    wrongAttempts = 0; // reset wrong count
  }
  else {
    wrongAttempts++;
    Serial.println("Access Denied ❌");

    if (wrongAttempts >= 3) {
      Serial.println("Too many wrong attempts! System locked for 30 seconds ⏳");
      digitalWrite(buzzer, HIGH);
      delay(3000);
      digitalWrite(buzzer, LOW);
      delay(30000); // lock period
      wrongAttempts = 0;
      Serial.println("You can try again.");
    }
  }
}

// --- Unlock door (servo) ---
void unlockDoor() {
  lockServo.write(90);    // Unlock position
  digitalWrite(buzzer, HIGH);
  delay(200);             // short beep
  digitalWrite(buzzer, LOW);
  delay(5000);            // door stays open for 5 seconds
  lockServo.write(0);     // lock again
  Serial.println("Door locked again 🔒");
}
```

- **Summary**

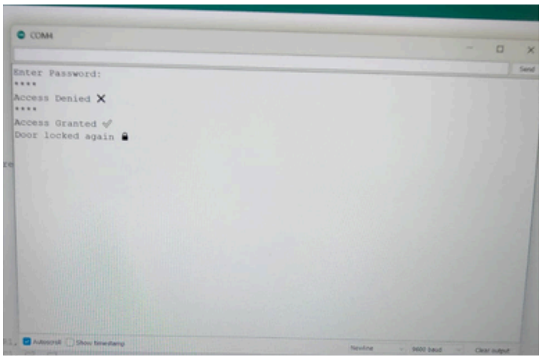
The Arduino acts as the control unit, reading keypad inputs, comparing passwords, and operating the servo and buzzer accordingly.
This integration of hardware and software creates a secure and automated digital lock system that demonstrates real-time password verification.

CHAPTER 6 – RESULT

The Digital Lock System project was successfully designed, implemented, and tested using the Arduino Nano.

When the correct password was entered through the keypad, the servo motor rotated, simulating the unlocking of a door or lock mechanism.
If the wrong password was entered three times consecutively, the buzzer activated and the system was locked for 30 seconds, preventing any further password attempts during that time.
The system worked accurately and responded quickly to keypad inputs.
It demonstrated how digital signals can be used effectively for security applications, using simple and cost-efficient components.

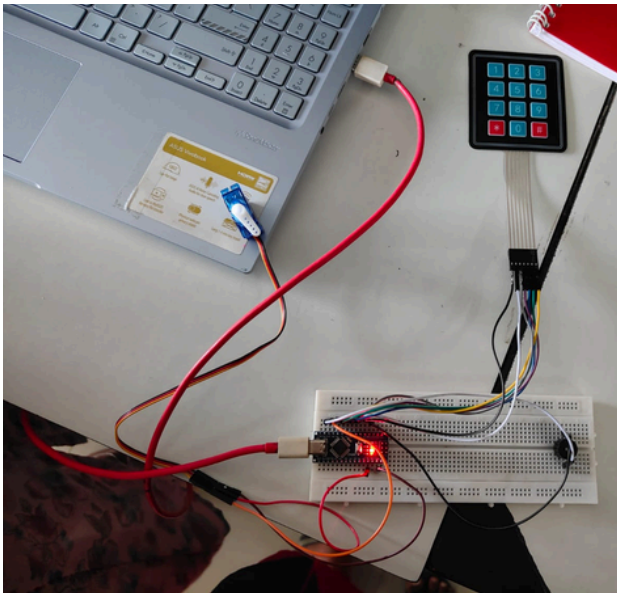
- **Observed Output**



Condition	System Response
Correct Password	Servo Motor Rotates (Lock Opens)
Wrong Password (1–2 times)	No Action / Retry Allowed
Wrong Password (3rd time)	Buzzer ON + 30 sec Lock
After 30 seconds	System Resets for Next Input

- **Overall Result**
    - The project proved successful in achieving its goals of:
    - Password-based authentication,
    - Controlled access,
    - Unauthorized attempt detection, and
    - Automatic lockout period after failed attempts.

CHAPTER 7
CONCLUSION



The Digital Lock System project provides a simple, reliable, and cost-effective way to enhance security using password-based access control.
It successfully demonstrates how digital signals and Arduino programming can be used to design an efficient locking mechanism without the need for mechanical keys.

By entering the correct password, access is granted through the servo motor acting as the lock.
If the password is entered incorrectly three times, the buzzer alerts unauthorized access, and the system locks for 30 seconds, ensuring safety from repeated attempts.

This project helps in understanding the concepts of digital input/output, password verification, and timing control in digital communication systems.
It can be further improved by adding an LCD display, wireless access, or IoT-based monitoring for enhanced functionality.